

Additional AVL Trees Problems

Filename: No File Submission

Instructions

This worksheet contains additional practice problems on AVL trees. Try to complete each problem on your own before looking at the solutions. The solutions will be provided separately, so make sure to give each problem an honest attempt first. Checking the answer too early defeats the purpose of the practice. As you work, time yourself and do not spend more than 10 minutes on any single problem.

TODO 1

Insert the following elements into an initially empty AVL tree in the given order. Draw the state of the tree after each of the underlined elements is inserted (i.e., insert, rebalance, if necessary, then show the state).

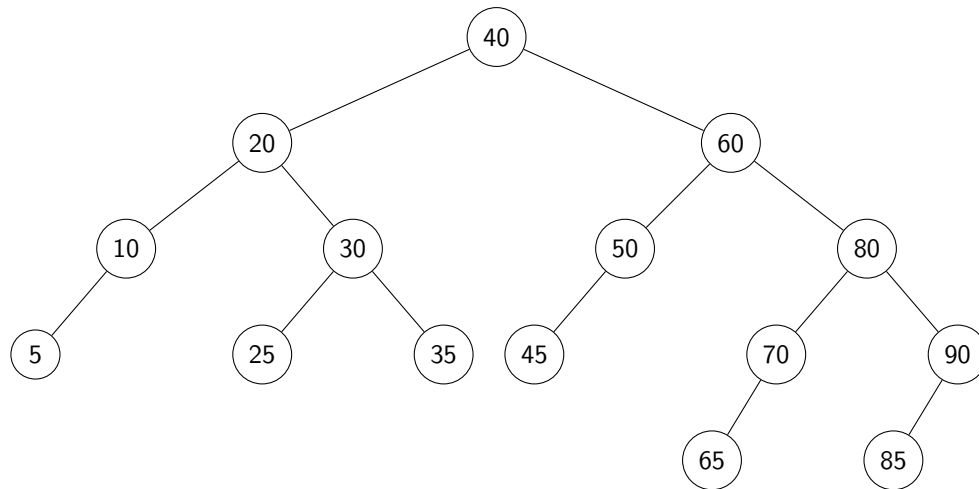
50, 80, 70, 99, 85, 100, 95, 84

After 70	After 85	After 100	After 95	After 84

Hint: Insert all elements into the AVL tree in the given order. However, you are only required to draw the state of the tree after inserting the underlined values, *after any necessary rebalancing has been completed*.

TODO 2

Consider the following AVL tree.



Delete the specified nodes. After each deletion, draw the resulting AVL tree *after any necessary rebalancing has been completed*.

5, 10, 20, 25, 30